

Justin Amen N'GBLOGNI

Research Internship: Embodied AI, Perception and Robotics (M.Eng. Year 5) | 6 March to 6 September 2027

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PROFESSIONAL SUMMARY

Final-year engineering student (M.Eng., ENSIL-ENSCI / ISAE-Supméca) specializing in **perception and control for autonomous systems**. At Politecnico di Torino I built a complete robotic navigation stack, from embedded semantic perception to GPU-accelerated predictive control, and carried it from mathematical modeling through to validation on physical hardware. I am looking for a 6-month final-year research internship in embodied AI, perception or control, ideally within a research team where the internship can lead to a PhD. Open to international relocation.

EDUCATION

M.Eng. Year 5 exchange programme in Mechatronics and Complex Systems <i>ISAE-Supméca, Graduate School of Mechanical Engineering</i>	2026 - 2027 <i>Paris, France</i>
M.Eng. in Mechatronics, Robotics and Automation (5-year integrated program) <i>ENSIL-ENSCI, Graduate School of Engineering (French Grande Ecole)</i>	2024 - 2027 <i>Limoges, France</i>
Integrated Preparatory Program in Electrical Engineering <i>ENSIL-ENSCI (France) / Ecole Polytechnique d'Abomey-Calavi (Benin)</i>	2021 - 2024 <i>France / Benin</i>
Scientific Baccalaureate, Highest Honors (Mention Très Bien) <i>St Augustin Secondary School</i>	2021 <i>Benin</i>

PROFESSIONAL EXPERIENCE

R&D Intern in Perception and Risk-Aware Autonomous Navigation <i>Politecnico di Torino, DIMEAS Department (advisors: Prof. F. F. Lizzio, Prof. E. Capello)</i>	May 2026 - July 2026 <i>Turin, Italy</i>
- Built an end-to-end navigation stack for a TurtleBot 4: a two-layer risk map (RGB-D semantic segmentation + geometric depth) driving a GPU-accelerated MPPI controller evaluating 5,120 trajectories per cycle at 14 Hz. - Results: cut risk exposure by 44% under forced traversal and by 100% in avoidable zones, at only 2-9% overhead in distance and travel time. Benchmarked and selected YOLO26 for embedded deployment (approx. 46 FPS on-device vs. 7-10 FPS CPU-only). - Derived a cost-arbitration model predicting the exact threshold at which the robot switches from traversing to circumnavigating a hazard; validated against physical measurements, then corrected and re-validated. Stack: ROS 2 Jazzy, Gazebo Harmonic, C++/CUDA, Python, PyTorch.	
Assembly Line Operator Intern <i>Stellantis Poissy, Stellantis Group (formerly PSA)</i>	June 2025 - July 2025 <i>Poissy, France</i>
Final assembly of Opel Mokka and DS 3 Crossback vehicles: hands-on exposure to takt time, quality standards, and safety protocols on a high-volume production line.	
Robotics & AI Instructor <i>Fondation Vallet (Fondation de France), EEIA Summer School</i>	2022 - 2025 <i>Benin</i>
Designed and led hands-on workshops in AI and embedded robotics for audiences ranging from middle-school students to working professionals.	

TECHNICAL PROJECTS

FlyRenov: redesign of a servo-controlled hose winder for a roof-cleaning drone <i>Industrial project, ENSIL-ENSCI x FlyRenov</i>	2025 - 2026 <i>France</i>
- Full machine redesign with no loss of functionality: 3x lighter (40 to 14 kg / 88 to 31 lb, now single-person portable), 4 motors reduced to 2, and an entire CAN network replaced by a single ESP32 board. - Closed-loop tension and length control of a 60 m tethered hose (VESC/FOC, UART); DFMA-oriented design, CNC machining, digital mock-up in 3DEXPERIENCE.	
Autonomous robotic arm playing tic-tac-toe via reinforcement learning <i>EEIA Summer School</i>	2025 <i>Benin</i>
Deep reinforcement learning agent (DQN/PPO) reaching 98% accuracy on complete games, with YOLO-based visual board detection and real-time inference embedded on a Raspberry Pi 5.	
Autonomous vehicle, from kit to ground-up design <i>EEIA Summer School</i>	2021 - 2024 <i>Benin</i>
From-scratch redesign of the vehicle (CAD in Fusion 360, 3D printing, electronics, software) with self-driving via imitation learning, from distance-sensor guidance to camera vision (TensorFlow, OpenCV, TinyML).	

TECHNICAL SKILLS

AI & Computer Vision: PyTorch, TensorFlow, OpenCV, semantic segmentation (SegFormer, MobileNet), YOLO, reinforcement learning, TinyML.	Programming & Embedded: Python, C, C++, CUDA, Arduino, ESP32, STM32, RTOS, Raspberry Pi, embedded Linux, AI-assisted development (LLMs). Protocols: CAN, SPI, I2C, UART.
Robotics & Perception: ROS 2 (Jazzy), Gazebo Harmonic, NVIDIA Isaac Sim, MPPI & model predictive control, SLAM, path planning, Kalman filtering, sensors (IMU, GPS, LiDAR, OAK-D RGB-D camera).	Design & R&D Tools: MATLAB/Simulink, Simcenter Amesim, SolidWorks, Fusion 360, 3DEXPERIENCE, EasyEDA, Altium Designer, KiCad, Git/GitLab.

AWARDS, LANGUAGES & INTERESTS

Awards & Certifications: Fondation Vallet Excellence Scholarship (2023-2027) • 2nd National Prize, EEIA (2021) • MathWorks (MATLAB, Simulink) • Arduino
Languages: French (Native) • English (Intermediate, CEFR B1) • Italian (Intermediate, CEFR B1) • German (Beginner, A1)
Interests: Automotive, Robotics & AI, Travel, Drawing, Music, Cooking.